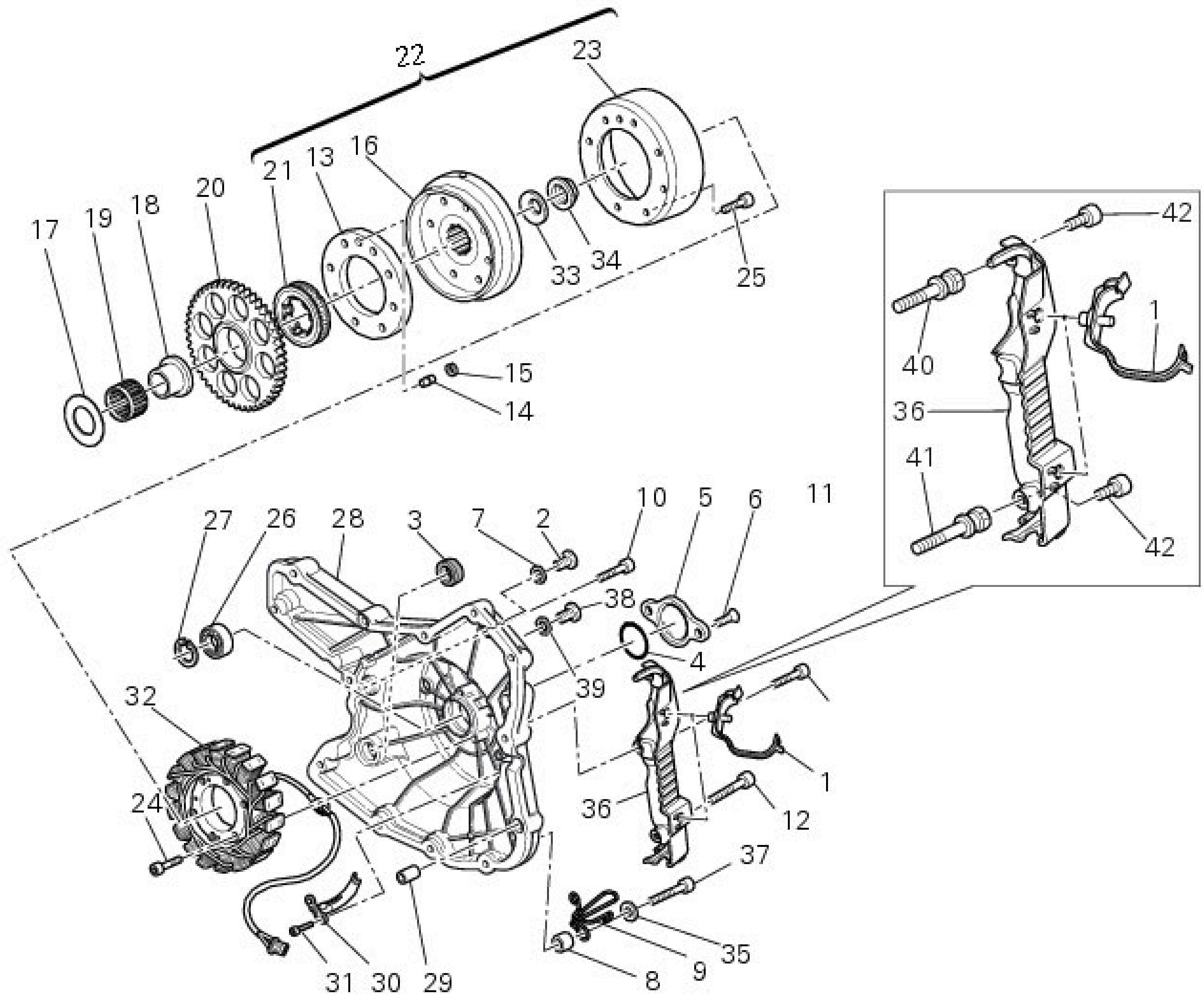
8 -Flywheel - alternator



1 Hose clip
2 Pick-up sensor inspection plug
3 Sealing ring
4 O-ring
5 Cover
6 Screw
7 Washer
8 Spacer
9 Cable guide bracket
10 Screw
11 Screw
12 Screw
13 Flange
14 Locating dowel
15 Circlip
16 Flywheel
17 Washer
18 Inner ring
19 Needle roller bearing
20 Electric starter driven gear
21 Starter clutch
22 Flywheel-starter clutch assembly
23 Alternator rotor
24 Screw
25 Screw
26 Bearing

27 Circlip
28 Generator cover
29 Locating bush
30 Bracket
31 Screw
32 Alternator stator
33 Belleville washer
34 Flanged nut
35 Washer
36 Breather hose cover
37 Screw
38 Plug
39 Aluminium gasket
40 Banjo bolt
41 Banjo bolt
42 Screw

Spare parts catalogue

1100 ABS	ALTERNATOR-SIDE CRANKCASE COVER
1100 ABS	ELECTRIC STARTING AND IGNITION
1100 ABS	AIR INTAKE - OIL BREATHER
1100 S ABS	ALTERNATOR-SIDE CRANKCASE COVER
1100 S ABS	ELECTRIC STARTING AND IGNITION
1100 S ABS	AIR INTAKE - OIL BREATHER

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

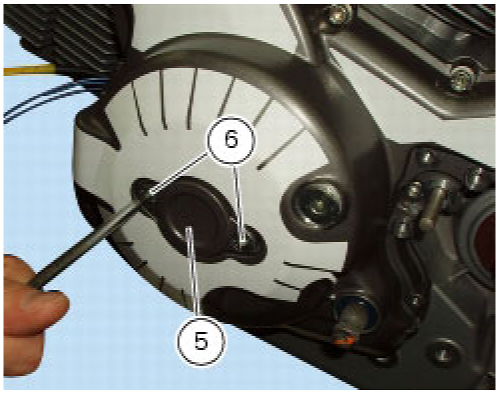
Removal of the generator cover

Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the gearchange control	F 5, Removal of the gearchange control
Remove the clutch slave cylinder	F 2, Removal of the clutch slave cylinder
Remove the front sprocket cover	G 8, Removing of the front sprocket

Note

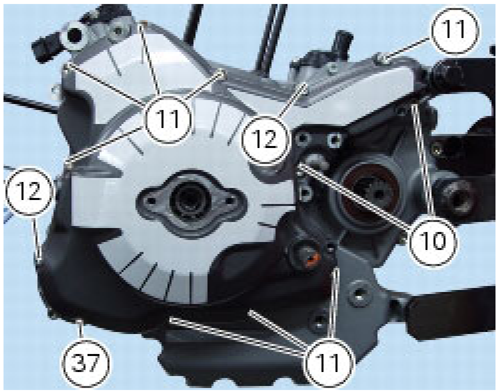
For clarity, the figures show the engine block removed from the frame.

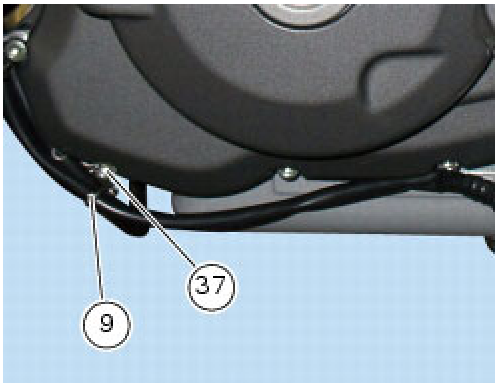
Unscrew the two retaining screws (6) of the centre cap (5) over the end of the crankshaft.

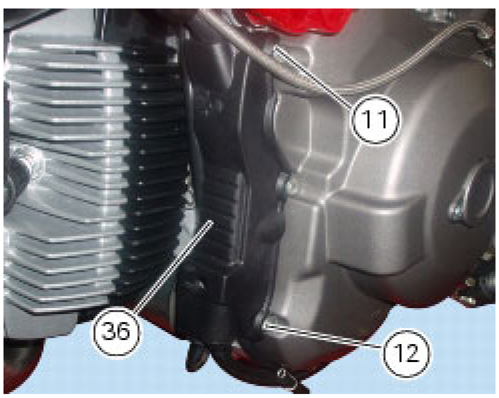


Unscrew the screws (10), (11), (12) and (37) securing the generator cover and cover (36).

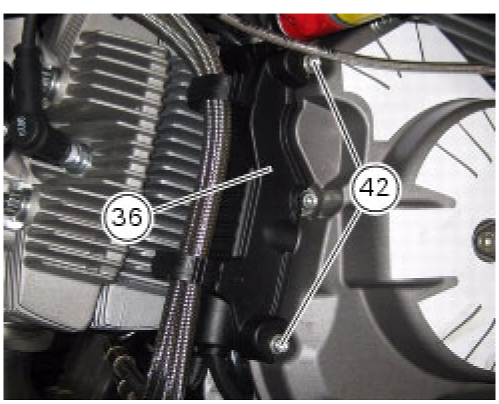
Recover the spacer (**8**), the Belleville washer (**35**) and the cable guide (9).



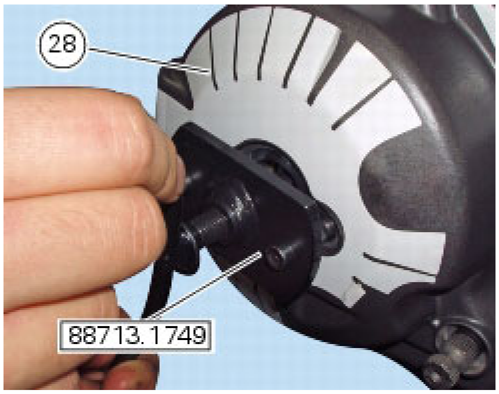




In some models you might find a different type of retaining screws for cover (36), as follows: there are two screws (42) retaining cover (36) to generator cover that are on top of the special screws (40) and (41) that fix the generator cover.

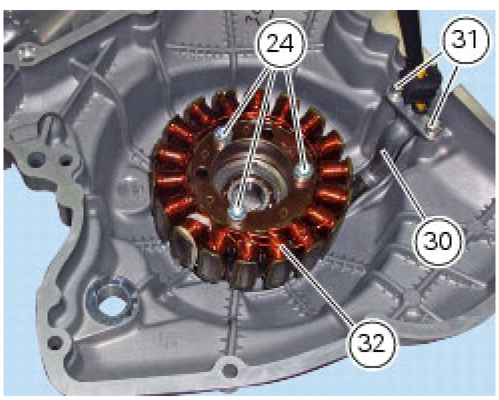


Fix service tool no. **88713.1749** to the holes left vacant by the two bolts (6) you have just removed.
Turn the tool shaft slowly to separate the cover (28) from the LH crankcase half.

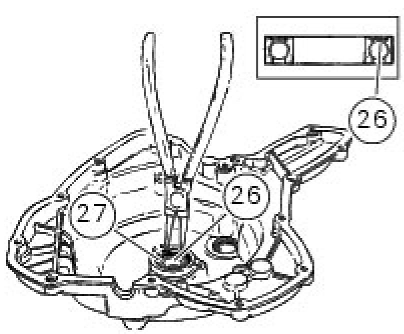


Disassembly of the generator cover

Undo the three stator retaining screws (24) and the two screws (31) securing the cable guide bracket (30) from inside the generator cover.
Remove the stator (32) and the cable guide bracket (30).



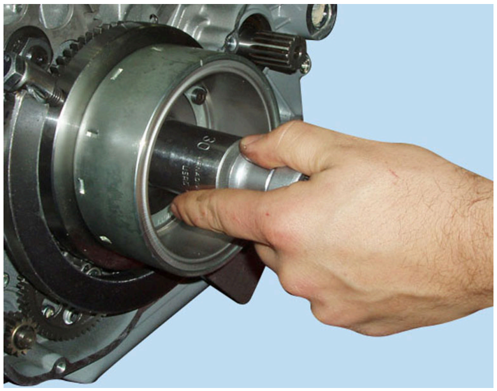
The generator cover is fitted with a bearing (26), held in place by circlip (27), which locates on the end of the crankshaft.
Remove the circlip (27) with circlip pliers.
Remove the bearing (26) using a universal puller.



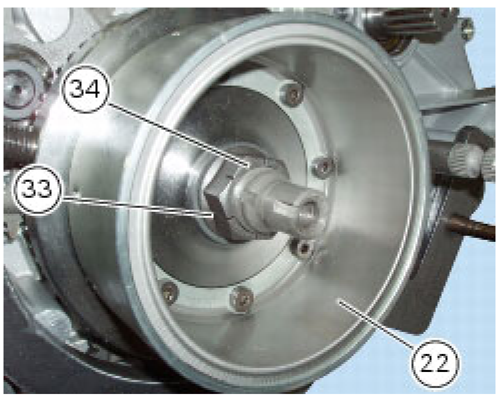
Removal of the flywheel/alternator assembly

Use service tool no. **88713.3367**.
Tighten tool to the flywheel.
Unscrew the alternator/flywheel nut (34), heating it with a hot air gun; do not use a naked flame as this could damage the starting system components.

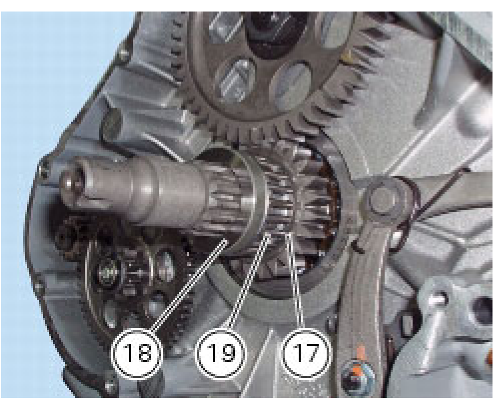
Warning
While unscrewing the nut, apply axial pressure to the socket to avoid damage or injury in the event of the wrench suddenly slipping off the nut.



Remove the nut (34), the Belleville washer (33) and the flywheel assembly (22) together with gear (20).



Remove the inner race (18), the needle roller bearing (19) and the washer (17).



Important
Examine the inner race (18), needle roller bearing (19) and internal washer (17) for wear.

Note
The hole in the inner race (18) allows the passage of oil for lubrication of needle roller bearing (19).

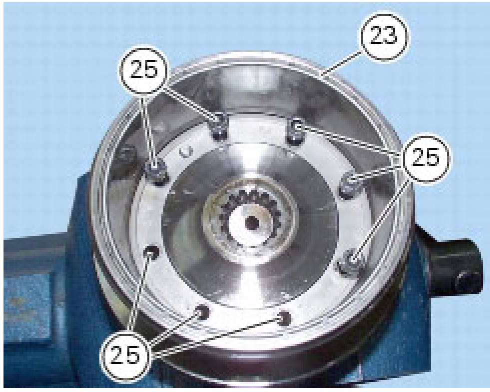
Checking the flywheel/alternator assembly

Examine the inner part of alternator rotor (23) for signs of damage.

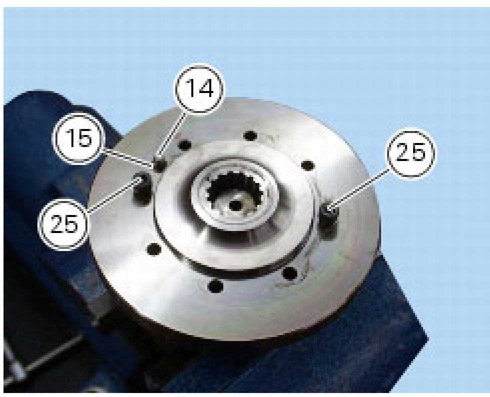
Check that the starter clutch is working properly and that the needle races do not show signs of wear or damage of any kind.
If there is any malfunction, remove the whole assembly.

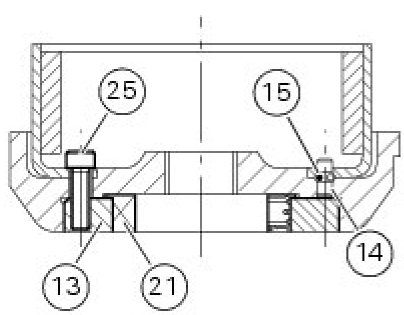
Disassembly of the flywheel-alternator assembly.

Unscrew the eight screws (25) and remove the alternator rotor (23) from the flywheel.



Slide out locating dowel (14) and circlip (15).
Insert two of the screws (25) just removed from flywheel rotor-side in their holes in order to remove the flange (13) and the starter clutch (21).
The starter clutch is a slight interference fit on the flange. To remove it, use a suitable drift.

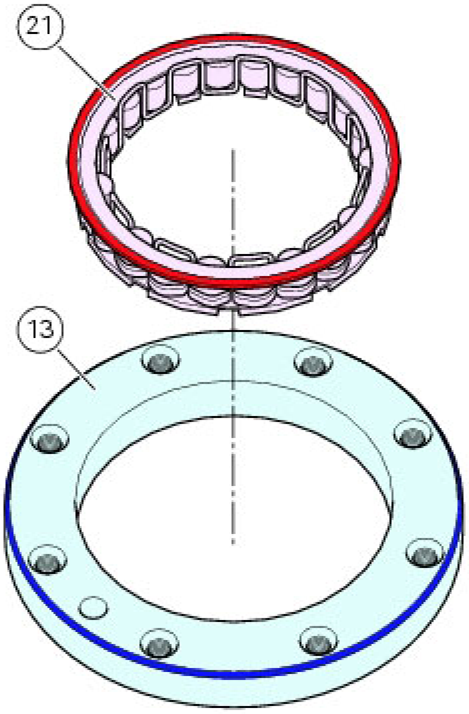


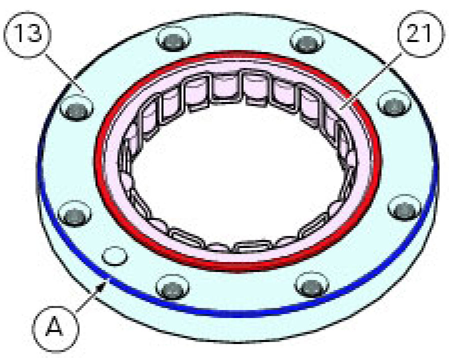


Reassembly of the flywheel-alternator assembly

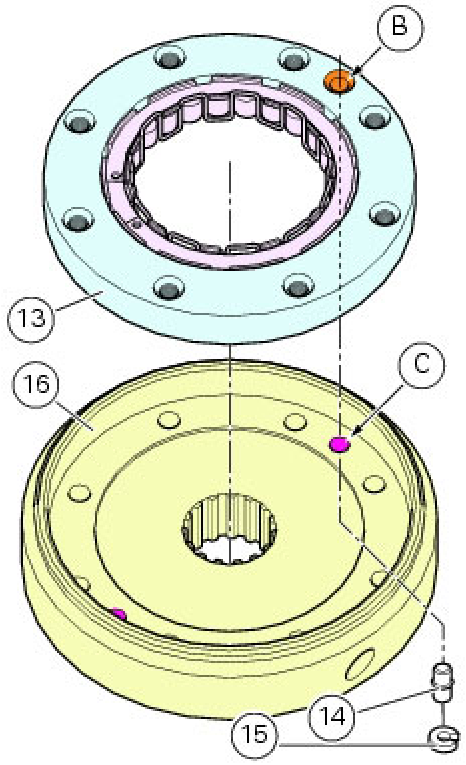
Install the starter clutch (21) in the flange (13) and take it fully home.

Important
Orient the flange with the round edge side (A) facing the starter clutch.



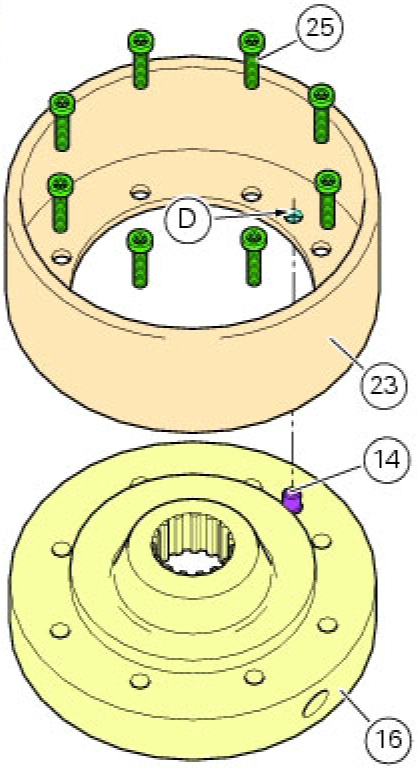


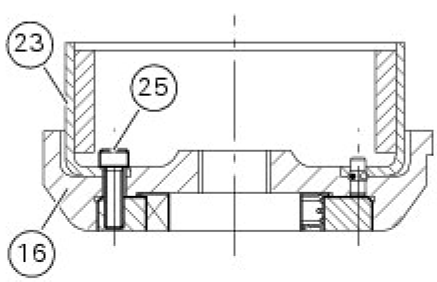
Seat the flange (13) with the starter clutch in the flywheel (16), aligning the flange locating hole (B) with the flywheel locating hole (C).
When refitting the flange to flywheel, insert a pin or a wrench inside the hole (C) for the rotor flywheel locating dowel (14) in order to line up the flywheel holes with flange threads. This is a useful tip, as the holes cannot be aligned after having fitted the flange to the flywheel due to the interference fit.
Orient the flange with the round edge side facing the flywheel.
Fit the locating dowel (14) with the circlip (15) in the flywheel locating hole (C).



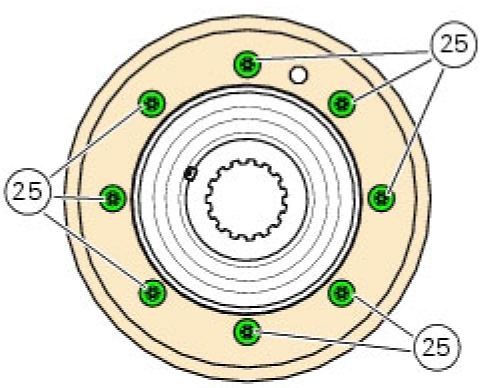
Install the rotor (23) on the flywheel (16), aligning the flywheel locating dowel (14) with the rotor locating hole (D).

Apply threadlocker to the rotor-flywheel fixing screws (25) and start them in their threads.

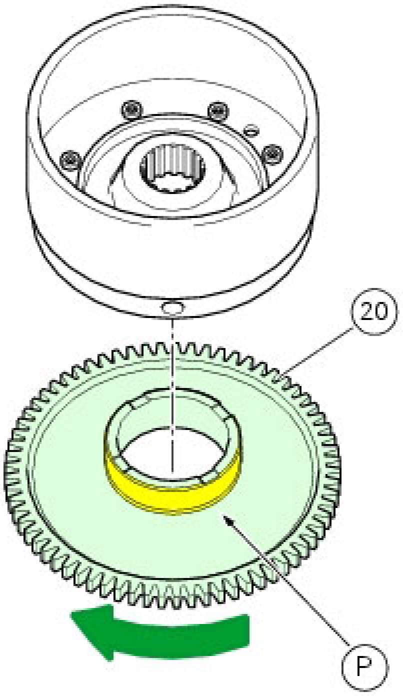




Tighten the screws (25) to the specified torque (Sect. C 3, Engine torque settings).

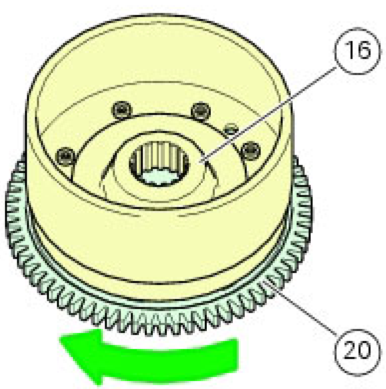


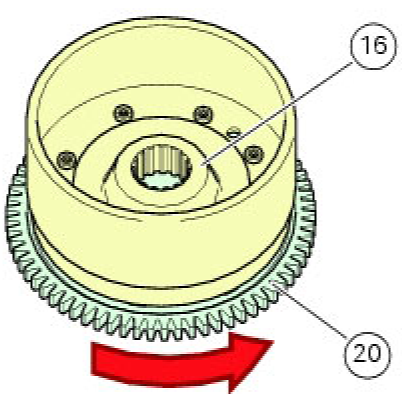
Lubricate the plate (P) of the driven gear (20) with engine oil.



Install the driven gear (20) on the starter clutch, ensuring it is properly seated.

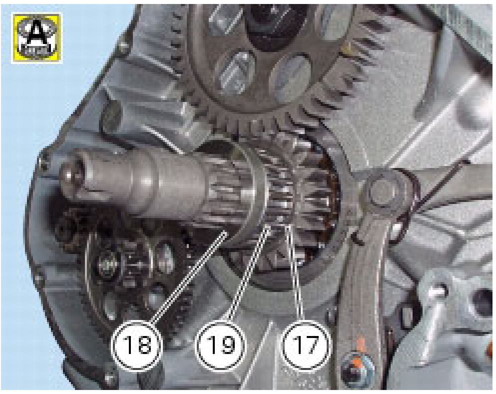
Check that the driven gear can rotate freely in the direction of the green arrow but not in the direction of the red arrow.
If either of these two conditions is not met, this means that the starter clutch has not been installed correctly.





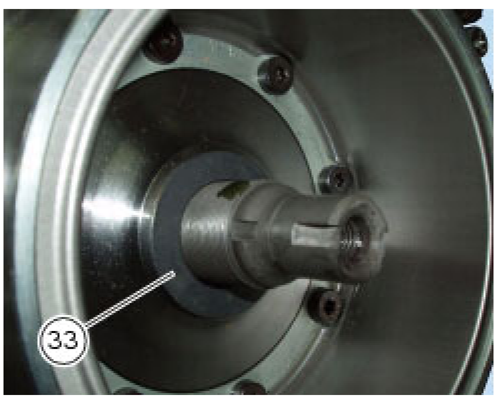
Refitting the flywheel-alternator assembly

Fit the washer (17), duly lubricated, the needle roller bearing (19), and the inner ring (18) to the crankshaft.
Make sure the inner ring is centred on the washer.



Fit the previously assembled flywheel-rotor-driven gear on the crankshaft, taking care not to alter the position of inner ring (18) with respect to the washer (17).
The flywheel reference mark must be aligned with the groove on the crankshaft, in correspondence with the keyway.

Lubricate the contact surfaces of Belleville washer (33) with engine oil.
Fit the Belleville washer (33) on the end of the crankshaft.



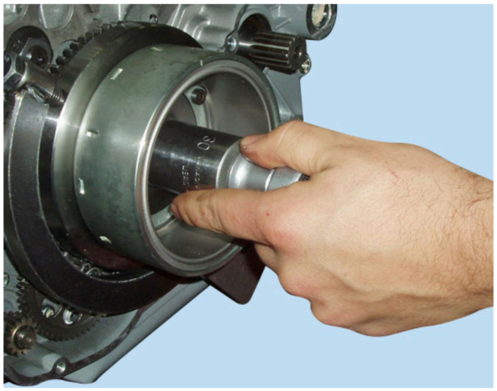
Apply recommended threadlocker to the crankshaft thread and to the flywheel nut thread (34).

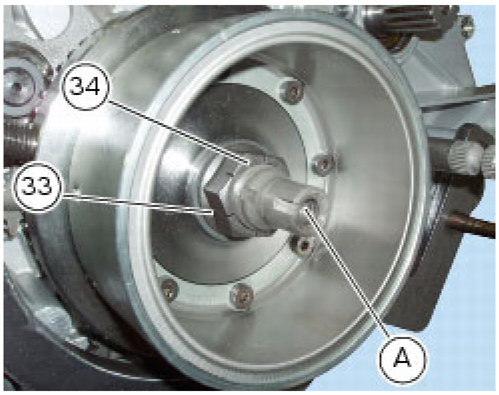
Restrain the rotation of the flywheel with the holding tool **88713.3367**.

Important

Tighten the nut (34) to the specified torque (Sect. C 3, Engine torque settings).

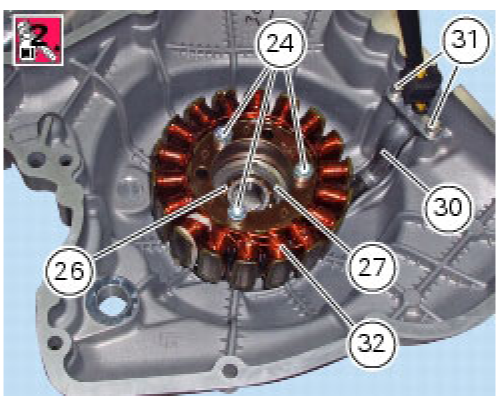
Remove all traces of Loctite from the threads of the nut (34) and the crankshaft (A).





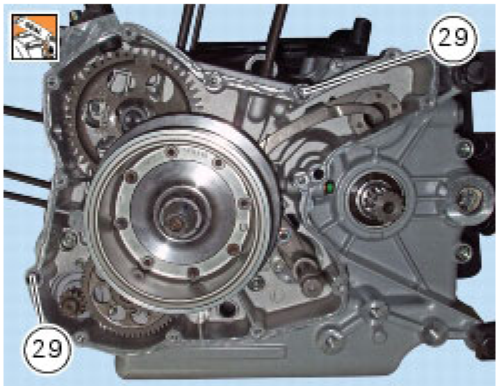
Reassembly of the generator cover

Fit the stator (32) to the generator cover, routing the cable through the notch in the cover.
Smear the stator screws (24) with recommended threadlocker and tighten to the specified torque (Sect. C 3, Engine torque settings).
Fit the cable plate (30) and fix it to the cover with the two screws (31).
Before refitting, make sure that the crankshaft end bearing (26) and relative retaining ring (27) are installed on the generator cover (28).

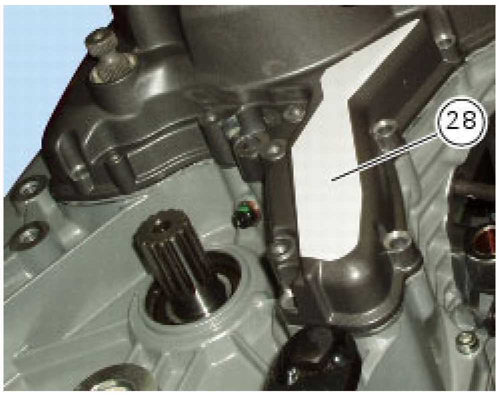


Refitting the generator cover

Remove any scale and grease from the mating surfaces of the left-hand crankcase half and the generator cover.
Fit the two locating bushes (29).



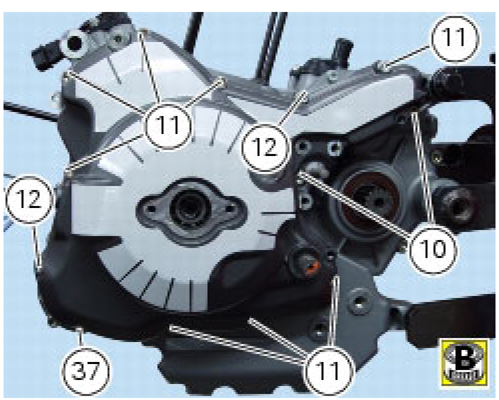
Spread a continuous uniform bead of DUCATI liquid gasket on the mating surface of the cover (28), ensuring continuity around the holes for the retaining screws and locating bushes.
Tap the cover at different positions with a rubber mallet to facilitate its location on the shafts and locating bushes.

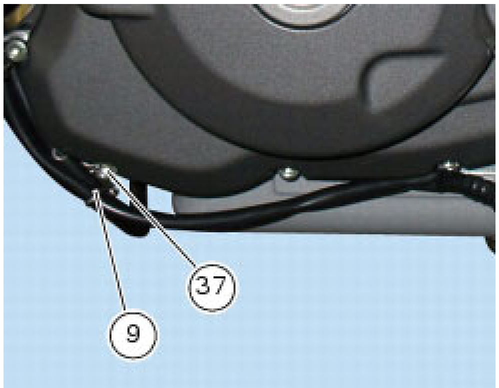


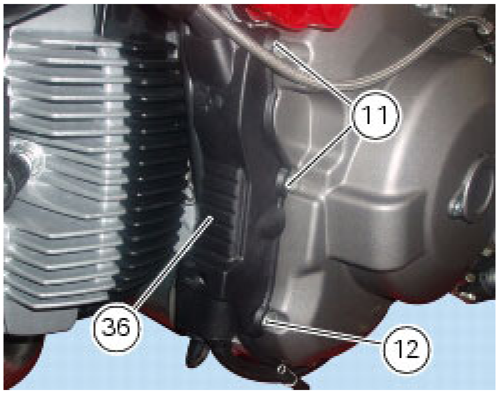
Insert the fixing screws in their holes following the indications given in the table.

Ref	Qty	Description
10	2	M6x20 mm screws
11	8	M6x25 mm screws
12	2	M6x30 mm screws
37	1	M6x35 mm screws

Install under the screw (12), in correspondence with the starter motor, the spacer (**8**), the guide (9), the washer (**35**) and install the cover (36) under screw (11) in correspondence with the alternator lead: for instructions on the positioning of the wiring under the cover (36) see below.
Tighten the fixing screws to the specified torque (Sect. C 3, Engine torque settings).

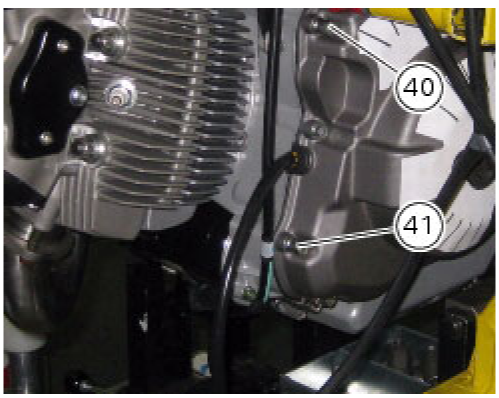


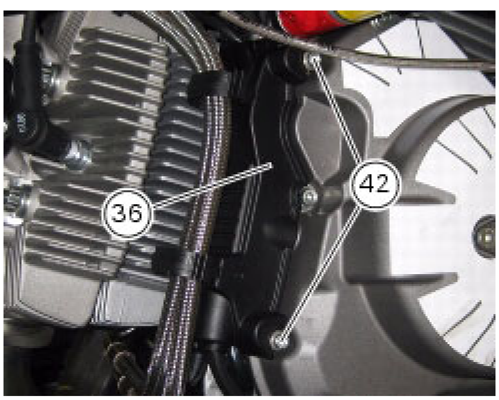




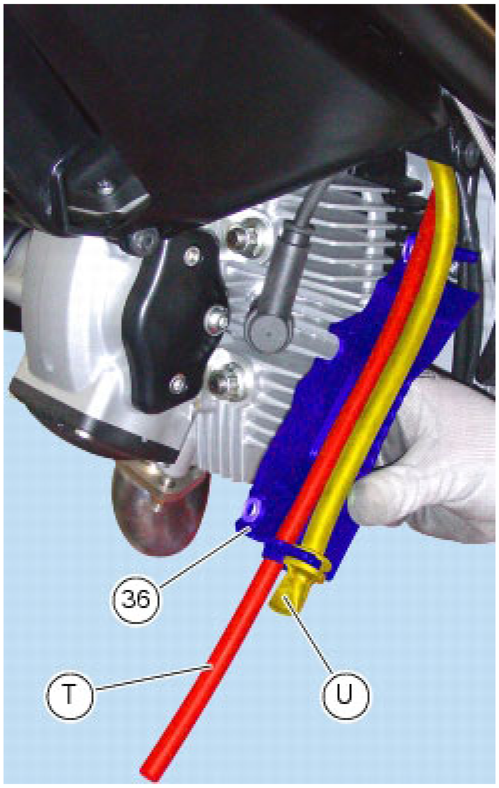
In some models you might find a different type of retaining screws for cover (36), as follows: there are two screws (42) retaining cover (36) to generator cover that are on top of the special screws (40) and (41) that fix the generator cover.

Fasten the generator cover tightening the special screws (40) and (41) (M6x30 mm) to the specified tightening torque (Sect. C 3, Engine torque settings) then fasten cover (36) to generator cover tightening the screws (42) to the specified torque (Sect. C 3, Frame torque settings).





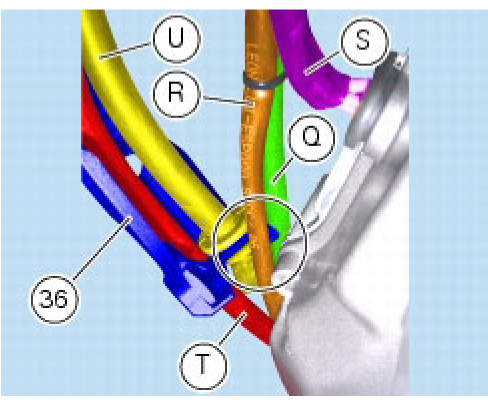
Check that the fuel tank drain hose (T) and the airbox drain hose (U) are positioned in the lower slot of the cover (36) as shown.

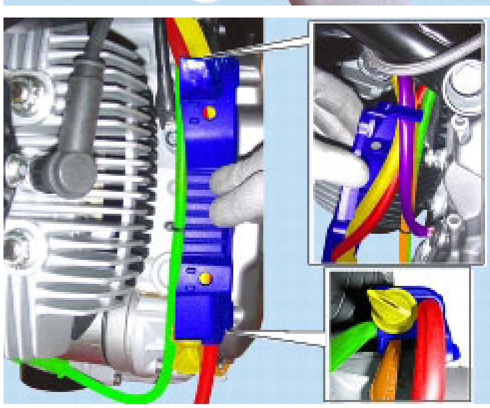


Offer the cover (36) up to the generator cover and position the fuel tank breather hose (T) and the airbox drain hose (U), the side stand cable (R), generator cable (S) and the starter motor cable (Q) as shown.

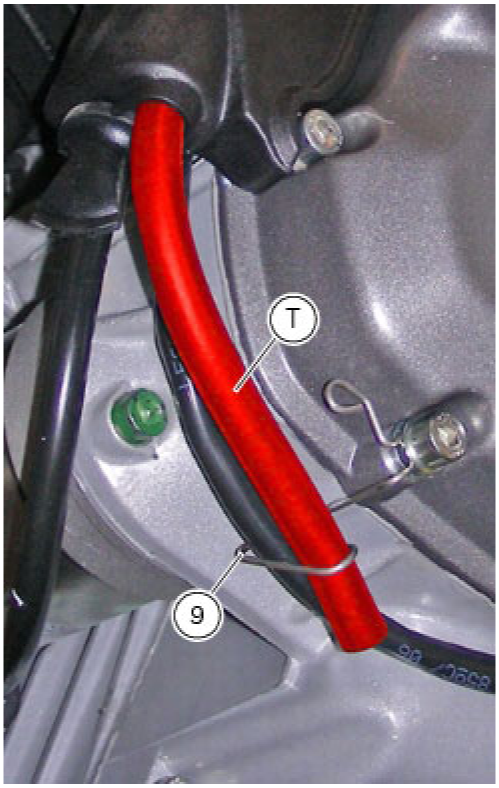
Important

The starter motor cable (Q) is the only cable that should be routed outside the cover (36).



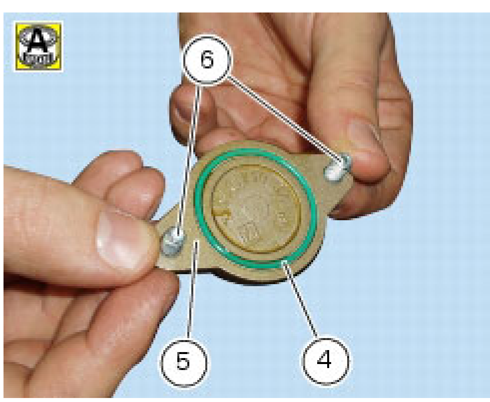


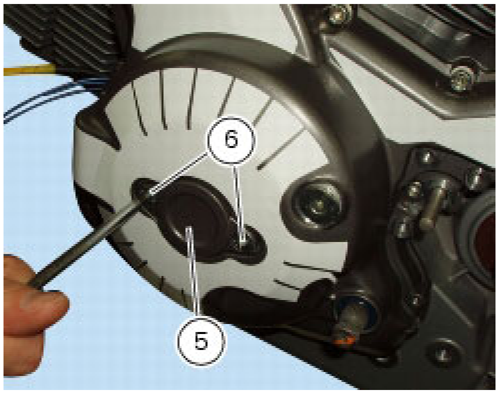
Position the fuel tank drain hose (T), inserting it in the guide (9).



Damp the sealing ring (3) with alcohol and install it in the generator cover, in correspondence with the gearchange shaft (Sect. F 5, Refitting the gearchange mechanism) by means of the suitable drift.

Make sure that the O-ring (4) is installed in the cover (5).
Apply threadlocker to the screws (6).
Tighten the two fixing screws (6) of the cover (5) matching the crankshaft to the specified torque (Sect. C 3, Engine torque settings).





Operation	Section reference
Refit the sprocket cover	G 8, Refitting the front sprocket
Refit the clutch slave cylinder	F 2, Refitting the clutch slave cylinder
Refit the gearchange control	F 5, Refitting the gearchange mechanism
Top up the engine oil	D 4, Changing the engine oil and filter cartridge

Positioning of cables/hoses under the breather hose cover mounted on the generator cover